SAMPLE

Earth's Dynamic Systems: Atmospheric Circulation

How Air Moves Around the Planet

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Introduction

Stand outside on a still afternoon and it is hard to believe the air is going anywhere. It is. At every moment the atmosphere of the planet is running a planetary delivery service, hauling the Sun's heat from the equator, where it arrives in embarrassing surplus, toward the poles, which would otherwise freeze solid, returning water to the land as rain, spinning up the storms that rearrange the weather of the mid-latitudes every few days, and doing all of it with machinery so large that a single loop of it can take a month to carry a parcel of air around the hemisphere. This book is about that machinery: the great circulation of the atmosphere, how it was discovered, why it is built the way it is, and what it is doing now that the planet is warming.

The subject has a history with a nice shape to it, because the global wind system was one of the first pieces of nature that humanity mapped scientifically and one of the last to be fully explained. Sailors knew the trade winds centuries before anyone could say why they blew; the doldrums and the horse latitudes had names before they had reasons; and the three-cell picture of the global circulation that still decorates every textbook began as an eighteenth-century guess and survived a century of objections before the mid-twentieth century, with the first global observations, showed what the atmosphere was actually doing, which was both less tidy and more interesting than the guess.

The tour follows the logic of the machine rather than its history. The first chapters establish the problem the circulation solves: the tropics receive more energy than they radiate, the poles less, and the imbalance would, unchecked, roast the one and freeze the other within a few decades; the atmosphere and the ocean between them are the compensation service. Then the founding forces get their chapters: heat and buoyancy, which start the air moving; the planet's rotation, which bends every current into the curving patterns that make the map of the winds look the way it does; and the tropical engine room, the deep belt of thunderstorms near the equator, whose rising air drives the trades and whose dry descents make the great deserts. The middle chapters climb latitude by latitude: the jet streams, the storm tracks, the frontal cyclones of the middle latitudes, and the cold cells at the poles. The later chapters widen out: the ocean's half of the partnership, the year-to-year moods of the system, the technologies that finally measured a planet-sized flow, and the settled question of why the whole engine is only predictable for a week or two no matter how good the instruments become. The final chapters face the present: what a warming world is already doing to the circulation, and what that means for the weather that follows it.

A word about the neighbors on the shelf. This series carries companion volumes on the atmosphere itself, on weather, on the jet stream, on monsoons, on El Nino, on chaos, on the oceans, and on energy, and each of those treats its own slice of the wind machine in more detail than this book gives it. Here the wind machine itself is the subject, and where a chapter touches a sibling's specialty, the treatment stays brief and the deeper account lives in its own volume. No equations follow, and the reader is asked for nothing but a general willingness to watch air behave like a fluid, which it is, and to believe that the planet's biggest weather map is also its most elegant machine.

The atmosphere's proportions deserve a paragraph of awe before the engineering begins: nearly all of the air lies within a few tens of kilometers of the surface, a film thinner, proportionally, than the skin of an apple, and the weather of the planet is the traffic of that film. The circulation's actors are crammed into the bottom of the sky, the storms' tops a dozen kilometers up, the whole machine running in a layer a commuter aircraft crosses in minutes, the planet's winds as surface phenomena of a very slightly fuzzy ball, the scale that makes the map both possible and necessary, the weather of a world, conducted in a sheet of gas thinner than the paint on a globe.

Chapter 1: The Problem the Circulation Solves

Begin with a bookkeeping error the planet refuses to make. The Sun, on average, heats the tropics more than the equator-facing surface can radiate back to space, and the polar regions less than they lose; the tropics therefore run an energy surplus and the poles a deficit, and if nothing moved, the tropics would grow hotter every year and the poles colder. Nothing of the sort happens, because two fluids, the air above and the ocean below, carry the surplus toward the deficit continuously, and the observed near-stability of the planet's temperature pattern is the direct proof that the transport is working. The atmosphere supplies roughly the transport of heat that the ocean does not, with its share largest in the middle latitudes and the total load split between the two fluids in a partnership that later chapters describe.

The engine framing deserves a careful paragraph, because it is the standard picture and it is easy to over-read.

The atmosphere behaves, in the large, like a heat engine: it takes in heat at the warm end, near the tropical surface, does work on the air, moving it against friction and lifting it against gravity, and exhausts heat at the cold end, near the poles and, more importantly, high up, where the atmosphere radiates to space. The engine's efficiency, in the strict thermodynamic sense, is low, a few percent, which sounds like an insult until one compares the power involved: the wind's kinetic energy budget, dissipated by friction and continuously regenerated by heating, is a number that dwarfs the output of every human power plant combined by a factor with several zeros. The planet's winds are a feeble engine of absurd size, and the weather is its exhaust.

The working substance is dry air with a variable cargo of water vapor, and the cargo is where the subtlety lives. Water vapor is lighter than the dry air it displaces, so moist air is buoyant; when it rises and cools, its vapor condenses, releasing the heat of the process and warming the rising parcel, which drives it higher; and the condensate falls out as rain, returning water to the surface and heat to the air. The latent heat carried by water vapor, picked up over the warm oceans and released in tropical thunderstorms, is a major fraction of the total heat the circulation moves poleward, and the planet's rain belts mark the places where the cargo is delivered. Any account of why the wind blows that omits water has told half the story, and the tropical chapters of this book lean on the fact.

What actually sets the pattern is a competition between three tendencies, worth naming now because they recur in every chapter. Heat wants to make air rise over warm regions and sink over cold ones: this is the direct overturning the earliest theorists imagined, hot air ascending at the equator and descending at the poles, one great loop per hemisphere. Rotation wants to bend any moving current to the side, relentlessly, until in the extratropics the wind settles into a near-balance where the pressure differences and the turning push against each other and the flow runs along the lines of pressure rather than across them. And the planet's geometry, a rotating sphere heated at the middle, wants to organize the whole flow into latitudinal belts and a few deep overturning cells rather than one simple loop. The observed circulation is the compromise the three tendencies reach, and its shape, the trade winds, the westerlies, the jets, and the polar easterlies, is the truce line between them.

The scale of the thing deserves one closing paragraph, because the circulation is the largest coherent motion a human can experience directly. A parcel of air caught in the tropical trade winds crosses an ocean in weeks; a parcel lifted in the equatorial thunder belt can take a month to descend in the subtropics; a wave rolling around the mid-latitude jet needs days to cross a continent. The velocities are modest, the distances are planetary, and the pattern is stable enough that generations of sailors, and before them generations of voyagers, bet their lives on its reliability, which is the historical point that opens the next chapter: the wind map was drawn by people trying to get somewhere, long before anyone could explain it.

The bookkeeping that the engine runs on has been measured to the point where it can be stated in round numbers. The planet, on average, receives sunlight at a rate that, spread over the whole surface, is about a quarter of a kilowatt on every square meter; a fraction bounces back to space from clouds and ice and bright surfaces, the rest is absorbed, mostly at the ground and sea surface in the low latitudes. The tropics, through the year, take in tens of watts more per square meter than they radiate to space, and the poles radiate away tens of watts more than they take in, and the difference, added over the enormous areas involved, is the freight the circulation exists to carry: a heat flow comparable, in total, to many millions of the largest power plants running at full output, all of it doing nothing more glamorous than keeping the equator's thermometer from creeping and the poles' from plunging, season after season, century after century.

The smallest cells run the same machine at the scale of an afternoon, and they deserve their place before the global tour begins: the sea breeze, the daily circulation between land and water, the land warming faster than the sea by noon, the air over the shore rising as the cool marine air slides in beneath it, and the land breeze running the same cell in reverse after dark, the circulation in miniature, a Hadley cell with a working life of twelve hours. The mountain-and-valley pair runs the same logic on terrain, the valley air rising by day and draining down at night, and the city adds its own version, the heat island's breeze flowing inward toward the warm core, the machine faithful at every scale, the map of the planet legible in the wind at the beach.